Megh Lad

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Summary

Computer Science undergraduate specializing in machine learning, computer vision, object tracking, and anomaly detection. Experienced in building research-driven ML systems for aerial imagery, segmentation, optimization-based learning, and real-time tracking using Python, C++, PyTorch, OpenCV, and scikit-learn.

Experience

Machine Learning Intern Jul 2025 – Aug 2025
Centre for Development of Telematics (C-DOT)

- Studied machine learning and deep learning techniques for aerial image analysis using high-resolution imagery.
- Explored segmentation, anomaly detection, feature extraction, and evaluated model behavior across dense visual scenes.

Research Intern Jan 2025 – Apr 2025
iEduVibhu

- Conducted research on Brain-Computer Interface (BCI) systems, signal preprocessing, and ML-based neurological signal interpretation.
- Investigated feature extraction and pattern recognition methods while contributing to SDG17 AI research documentation.

Research

Physics-Guided ADMM for Hyperspectral Anomaly Detection

- Proposed OODA-ADMM, a physics-guided RPCA framework combining ADMM, spectral unmixing, and material-aware feedback.
- Improved mean AUC from 0.962 to 0.971 and achieved 0.906 AUC at 10% target abundance versus 0.849 for RPCA.
- Validated using benchmark datasets, sub-pixel injection experiments, and ablation studies.

Projects

GPS-Denied Cooperative UAV Swarm Localization (Python, GTSAM, ArduPilot SITL, MAVLink, NumPy)

- Developed a cooperative localization pipeline for GPS-denied multi-UAV swarms using factor graph optimization.
- Implemented distributed localization(ADMM),Kalman filtering, robust outlier rejection,incremental iSAM2 optimization.
- Integrated ArduPilot SITL and MAVLink to stream external vision pose estimates into the EKF.
- Evaluated localization performance under sensor noise, communication failures, and dynamic swarm trajectories.

CPU-Only Real-Time Aerial Object Tracker (C++17, OpenCV)

- Developed a real-time aerial object tracker from scratch using C++17 and OpenCV.
- Implemented MOSSE correlation filtering with Kalman-based motion prediction.
- Added confidence-gated updates, motion-based re-detection, and log-polar rotation compensation.
- Outperformed OpenCV CSRT/KCF on small fast targets and extended to multi-target identity-preserving tracking.

AI-Powered Smart Attendance System (Python, Flask, YOLOv8, OpenCV, MongoDB)

- Developed an AI-powered attendance platform using Python, Flask, YOLOv8, OpenCV, and MongoDB.
- Implemented face enrollment, facial embedding generation, and real-time identity recognition for automated attendance.
- Built secure student and teacher portals with REST APIs for authentication, attendance sessions, and record management.
- Integrated MongoDB Atlas to store student profiles, facial embeddings, and attendance history with cloud persistence.
- Automated classroom attendance using real-time video processing, reducing manual effort and improving record accuracy.

Skills

Languages: Python, C++, Java, C, JavaScript, SQL
ML/AI: PyTorch, scikit-learn, NumPy, Pandas, Machine Learning, Deep Learning, Model Evaluation
Computer Vision: OpenCV, Object Tracking, Image Segmentation, Feature Extraction, Anomaly Detection
Research: ADMM, RPCA, Spectral Unmixing, Hyperspectral Imaging, Statistical Analysis
Robotics & Estimation: GTSAM, Kalman Filtering, Factor Graph Optimization, MAVLink, ArduPilot SITL
Tools: Git, GitHub, VS Code

Education

B.E. Computer Science 2023 – 2027
BMS Institute of Technology and Management, Bangalore CGPA: 8.64

Certifications

Deep Learning and Computer Vision (A-Z) – Udemy
Artificial Intelligence A-Z 2024 – Udemy